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Details Questions

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Question 0.54 pts

DNA sequences that serve as binding sites for proteins regulating transcription and are often found adjacent to a gene are said to be

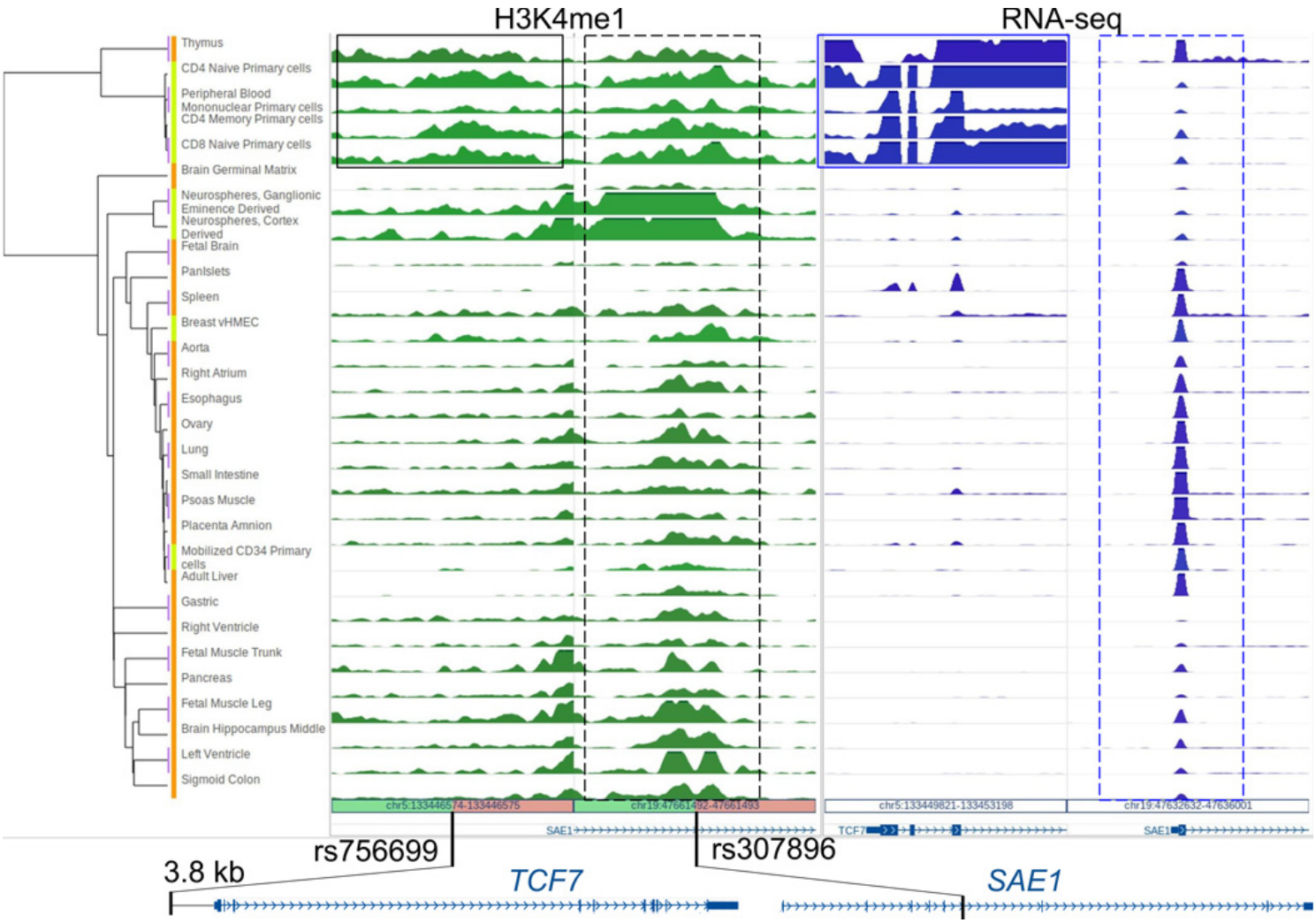
cross-reacting
cis-acting
trans-acting
origins of transcription
transcription factors

Question 0.55 pts

Epigenomics data can aid in interpreting the function of SNPs identified from GWAS studies. For example, two non-coding GWAS SNPs (rs756699 and rs307896) were found to be associated with multiple-sclerosis. Below is a genome browser image showing H3K4me1 ChIP-seq read density (in green) for a 6kb region centered on each SNP. H3K4me1 is a histone mark associated with enhancers. RNA-seq (RNA sequencing) read density (in blue) is shown over the 5' end of genes that are closest to these SNPs. RNA-Seq is a method to sequence mRNA. The closest gene to rs756699 is TCF7 and the rs307896 SNP is located in the intron of SAE1.

The epigenomic and expression data from 31 primary human tissues (orange) and cells (light green) are shown to the left. Hierarchical clustering (shown at the far left) is applied to both H3K4me1 and RNA-seq data to group together the tissues and cells that are most similar to each other based on the H3K4me1 and RNA-Seq data shown.

The enhancer associated with rs756699 seems to only be functioning in one type of tissue. Which one? (Hint: You can do an internet search on the tissue and cell line names for more information)



Immune

Brain

Lung

Colon

Question 0.54 pts

Do you think rs756699 is a causative mutation for multiple sclerosis?

Yes

No

Possibly, but additional experiments necessary

Question 0.54 pts

You are interested in performing a follow-up experiment for rs756699. By looking at the sequence around the rs756699 SNP, you predict that it could be a binding site for one of 5 transcription factors. Which experimental approach would help you determine which of the 5 transcription factors are binding this region?

ATAC-Seq

ChIP-Seq

Question 0.54 pts

The Epigenomics Roadmap had the data you needed to find that the transcription factor NF-kB binds directly to DNA that contains the SNP rs756699. Do you think this is a causative mutation for multiple sclerosis?

Yes
No
Possibly, but additional experiments are necessary

Question

Consider a eukaryotic gene regulatory pathway where a small molecule X activates the expression of a gene **C**. You have isolated recessive loss-of-function mutations in two different genes, **A** and **B**, both of which lead to uninducible expression of gene **C.** Genes **A** and **B** are not linked to each other and neither gene is linked to **C**.

Genes A and B encode transcription factors that act in a series. There are two possible orders in which these transcription factors act.

(1) X -> A -> B-> C

(2) X -> B -> A -> C

In order to distinguish between the two models above, you try to perform an epistasis test.

Question 0.55 pts

In order to perform the epistasis test, you isolate an allele of gene **A** that gives constitutive expression of gene **C**. This allele, called $\mathbf{A^S}$**,** causes a dominant phenotype of constitutive expression of gene **C**.

You perform the epistasis test by crossing a A^S/A^S B+/B+ male to a A+/A+ B-/B- female.

For model 1 above, what would be the expected phenotype in the F1?

constitutive expression of C
regulated expression of C
uninducible expression of C

Question 0.55 pts

For model 2 above, what would be the expected phenotype of F1?

constitutive expression of C
regulated expression of C
uninducible expression of C

Question 0.55 pts

Now you cross the F1 among themselves to produce F2. For model 1 above, give the expected ratio of constitutive, regulated, or uninducible phenotypes among the F2.

9:7:0
9:3:4
12:3:1
13:3:0

15:1:0

Question 0.55 pts

If the above F2 phenotypes were observed, what would be the type of genetic interaction?

Reciprocal recessive epistasis
Recessive epistasis
Dominant epistatsis

Question 0.54 pts

For model 2, what would be the expected ratio of constitutive, regulated, or uninducible phenotypes among the F2.

9:7:0
9:3:4
12:3:1
13:3:0
15:1:0

Question 0.55 pts

If the above F2 phenotypes were observed, what would be the type of genetic interaction?

Reciprocal recessive epistasis
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